

Quiz 1

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given a first-order differential equation

$$y' = e^{-3y}.$$

Verify that $y(x) = \frac{1}{3} \ln(3x + C)$ defines a one-parameter family of solutions of the given differential equation, then determine a value of the constant C so that $y(x)$ satisfies the initial condition $y(1) = 0$.

$$y(x) = \frac{1}{3} \ln(3x + C) \Rightarrow \begin{cases} y'(x) = \frac{1}{3} \cdot \frac{1}{3x+C} \cdot 3 = \frac{1}{3x+C} \\ e^{-3y} = e^{-3 \cdot \frac{1}{3} \ln(3x+C)} = e^{-\ln(3x+C)} = \frac{1}{3x+C} \end{cases}$$

$\Rightarrow y' = e^{-3y}$, and thus $y(x) = \frac{1}{3} \ln(3x + C)$ defines a one-parameter family of solutions for $y' = e^{-3y}$

$$\begin{aligned} y(1) = 0 &\Rightarrow \frac{1}{3} \ln(3 \cdot 1 + C) = 0 \Rightarrow \ln(3 + C) = 0 \\ &\Rightarrow 3 + C = 1 \Rightarrow \underline{C = -2} \end{aligned}$$

Problem 2. (5 points) Find a general solution for the first-order differential equation

$$\frac{dy}{dx} = \frac{1}{\sqrt[3]{2x+1}}.$$

$$\begin{aligned} y(x) &= \int \frac{1}{\sqrt[3]{2x+1}} dx = \int (2x+1)^{-\frac{1}{3}} dx \\ &= \frac{3}{2} (2x+1)^{\frac{2}{3}} \cdot \left(\frac{1}{2}\right) + C \\ &= \frac{3}{4} (2x+1)^{\frac{2}{3}} + C \end{aligned}$$